

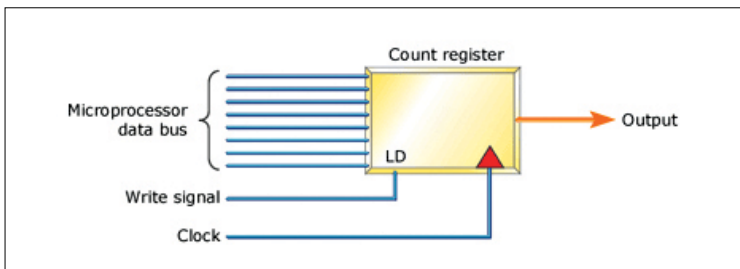
Timers

Timers are one of the most useful parts of any computing system. To measure the time taken between any two events is a crucial part of a lot of operations in the real world. We cannot imagine a scenario where timers do not make an impact. Timers can either count up (like a stopwatch) or count down (like a countdown timer). Keeping track of current time after the time has been set, measuring time taken for a task to be completed, cancelling a task when the task has been executing beyond expected length, starting a task at a given time are all done using timers in microcontrollers.

However, that is not the only use. One of the most important uses of a timer comes up in rather sophisticated SoCs which perform multi-tasking. An electronic timer can be used for raising interrupts on the hardware level. These interrupts are captured by the processor and an OS routine is invoked. This routine is called an interrupt handler. If the microprocessor of the SoC is a single-core processor, then the running application is first suspended (the execution is paused and the state is saved so that it can be resumed) before invoking the routine. This routine then runs multiple checks against the current state and decides what to do. A

NOTE

There is yet another use of a timer from a pure-hardware viewpoint which makes the timer the most important part of any chip – clock ticking. This is the “hertz” in the specification-sheet. A timer which counts very fast would be called one with higher frequency. All digital equipment depend on clock ticks of their respective chips. This tick is an electric signal produced inside the IC which tells the units to perform the next step. This ticking depends on a digital timer. Timers are thus part and parcel of almost all ICs.



That's how timers work